

From Digital Literacy to Career Adaptability: The Mediating Role of Employability

ABSTRACT

The basic competencies needed by modern companies have been drastically altered by the explosive growth of digital transformation and artificial intelligence. Technical expertise is insufficient in today's world. To succeed in a technology driven workplace, modern professionals need to have high digital literacy, agility in adopting new technologies, critical problem solving skills, and a commitment to lifelong learning. Even though digital tools are widely used in corporate and academic settings, many workers and graduates fall short of these changing standards. Inadequate preparation for highly digitalized work contexts and a lack of specialized digital skills are the causes of this gap. This study examines the direct relationship between digital competency and general workplace competitiveness in order to address this issue. This looks at how having excellent digital skills improves employment preparedness and fosters long term career sustainability in the contemporary economy. This study identifies a number of important scholarly gaps by a thorough systematic evaluation of the body of existing literature. These include contradictory theoretical frameworks, unclear empirical data about how digital literacy actually affects job outcomes, and a glaring lack of research on poor nations. The paper presents a comprehensive and integrated strategy to overcome these discrepancies. In order to provide a more accurate assessment of the existing workforce's preparation, this model combines digital competences, personal adaptability, and workforce readiness. In the end, this realization provides researchers, academic institutions, legislators, and business organizations with a crucial strategic framework for creating excellent digital training programs for the workforce of the future.

Keywords : *digital literacy, attitude towards digital literacy for technology, intention to use technology, career adaptability, employability*

1. INTRODUCTION

The massive acceleration of digital transformation accompanied by the adoption of artificial intelligence (AI) has fundamentally reconstructed the organization's operations, the labor market order, and the competency qualifications of the workforce at the global level. The corporate sector is increasingly integrating digital information systems, automation technology, cloud computing, data analysis, and AI-based

platforms to boost operational efficiency, innovation capacity, and long-term competitive advantage in the midst of a dynamic business environment (Lu et al., 2025). At the same time, the emergence of the era of Industry 4.0 and Society 5.0 increasingly strengthens the urgency of fulfilling human resources that are digitally responsive and capable of adapting to the ever-changing technology ecosystem. Today's modern work environment not only requires graduates of higher education institutions and workers to master the academic knowledge base, but also requires the ownership of digital literacy, technological flexibility, problem-solving acumen, communication skills, and continuous learning orientation in order to function optimally in a technology-intensive workspace (Vuorikari et al., 2022). A number of recent research also confirms that the ability to absorb labor (employability) in the digital era is increasingly determined by personal capabilities in uniting technological skills with cognitive skills and adaptive behavior (Singh et al., 2025).

The increasingly widespread penetration of digital technology in various industrial sectors has significantly changed the organization's expectations of work readiness and the competitiveness of prospective workers. Digital transformation is no longer limited to modifying the operational system, but also shifts the competence standards that must be met by workers to remain relevant in the modern labor market. Company management now prioritizes candidates who are able to operate digital technology efficiently, adaptive to system updates, able to collaborate in virtual space, and proactively respond to the innovation-driven business climate. Lambekova et al. (2025) argued that digital competence has become a strategic catalyst for economic reconstruction, a stimulant for organizational innovation, and business sustainability in the Sixth Information Revolution era. This shift indicates that a person's work absorption is no longer absolutely based on a diploma or formal education qualification, but on the capacity of workers to adapt continuously to the dynamics of the fluctuating digital ecosystem.

Although the urgency of digital competence in the modern work environment is increasing, the reality in the field still shows great challenges related to work readiness and optimization of employability levels. Higher education institutions take on a great responsibility to produce graduates who are ready to function in the midst of

an organizational ecosystem loaded with technology. However, in reality, many new graduates face major obstacles when going through the transition phase towards the world of work due to the lack of alignment between the quality of their skills and the real expectations of the industrial market. Empirical investigation by Mandal et al. (2026) shows that graduates from the Generation Z group often encounter work absorption constraints triggered by low technological flexibility, limited skills in solving digital problems, and mental unpreparedness in facing a digitally mediated work environment. In line with that, Nguyen (2026) found the fact that employers consistently complain about the existence of a competence gap in graduates, especially in relation to digital adaptability, interactive communication capabilities, and operational readiness in rapidly changing technology systems. This phenomenon confirms that the external education sector is often unable to keep up with the acceleration of competence needs from contemporary organizations.

The essence of this problem becomes increasingly crucial because the disruption of digital technology not only touches the technical aspects of work, but also penetrates the realm of organizational behavior, interaction patterns between employees, and the course of the long-term career development process. The workforce is required to simultaneously learn new digital systems, assimilate cutting-edge technology into work routines, and maintain adaptive capacity in the midst of uncertain business situations. Zhang (2025) confirmed that the mastery of digital literacy contributes positively to job quality and competitiveness in the labor market, because individuals with mature digital proficiency tend to have better occupational flexibility and wider access to job opportunities. On the other hand, digital competence deficit risks reducing the degree of employability, increasing the vulnerability of workers' positions in the job market, and widening the skill mismatch between the curriculum of educational institutions and the real needs of the industrial world.

In order to respond to this problem, many academics direct their research focus to the analysis of the relationship between digital literacy, technology adjustment patterns, and external work absorption. As an illustration, Testa et al. (2026) evaluated the implications of AI literacy as well as AI readiness towards the level of career adaptability in a group of business and management students. The results of their study

prove that the understanding of AI literacy contributes positively to the strengthening of career adaptability through the mechanism of building technological self-confidence and individual adaptive capacity. However, the study focuses on the aspect of readiness and flexibility of AI alone, and does not comprehensively review the dimensions of technology acceptance behavior and external employability. On the other hand, Imjai et al. (2025) found that the digital learning ecosystem and digital adaptation skills had a great influence on work-based digital competence in Generation Z students. This emphasizes that adaptive technology behavior plays a vital role in preparing graduates to enter a technology-based job market.

In addition, a number of previous studies also raised the importance of the involvement of psychological mechanisms and behavioral aspects in shaping the external work readiness in the digital environment. A study that adopts the Technology Acceptance Model (TAM) framework confirms that the orientation of a positive attitude towards digital devices accompanied by a strong behavioral intention to utilize the technology system significantly is able to optimize technology adoption, user attachment, and the effectiveness of the learning process in the academic field (Ahmed et al., 2021). Furthermore, Hasan et al. (2026) underlined that the digital growth mindset (digital growth mindset) along with adaptive technology behavior is able to strengthen entrepreneurial learning capacity as well as functional skills related to the employability of students in universities. The systematic review carried out by Lan et al. (2025) also revealed the fact that digital literacy significantly supports the course of the self-regulated learning process (self-regulated learning) as well as adaptive learning actions that are absolutely necessary for the maturation of sustainable work absorption in the digitized work scope.

Although the previous research lineup has presented valuable insights about digital competence and work absorption, there are still some theoretical and empirical gaps that have not been completely solved. First, most of the previous studies tend to dissect the constructs of digital literacy, technology acceptance, adaptability, and employability as separate entities, rather than integrating them into a whole analysis model. As a result, the understanding of how digital competence interacts simultaneously with the orientation of the attitude of using technology, the intention of

digital system adoption behavior, and career adaptation capabilities in determining the external absorption of work so far is still very limited. Second, there is an inconsistency in empirical findings regarding the direct and indirect effects of digital literacy on employability. One camp argues that digital proficiency is directly able to increase the value of employability because individuals who respond to automatic technology have high market competitiveness (Zhang, 2025). On the other hand, another camp stated that the external absorption of work is indirectly influenced by adaptive intermediate psychological variables, such as the level of career adaptability, technological confidence, self-efficacy, and the intention to use technology (Testa et al., 2026). Other important limitations stem from the contextual and methodological fragmentation of previous research, where the majority of studies take the locus in developed countries or examine the partial relationship between variables separately. The empirical evidence that validates the simultaneous relationship between digital skills, technology acceptance behavior, adaptation capacity, and work absorption capacity in the landscape of developing countries is still minimal, even though the acceleration of digital transformation in developing economies is accelerating rapidly. Moreover, a systematic analysis of work readiness interventions indicates that studies in the higher education sector still lack an integrated framework that is able to explain the formation of overall work absorption in a digitally disrupted organizational climate (Barachino et al., 2025). In line with that, Lan et al. (2025) emphasized that there is a real methodological fragmentation in digital literacy research, especially related to the unification of behavioral and adaptive variables into employability modeling.

To fill these scientific gaps, this research is realized to analyze the relationship between the digital skill capacity of individuals and their job absorption status, by integrating the perspective of digital competence, technology adaptation process, and employability framework into one comprehensive theoretical model. Through this integrative approach, this research seeks to provide a deeper conceptual understanding of the chain mechanism of how digital literacy capital transforms itself into a real external achievement of employability in the digitized labor market. From a practical point of view, this study is expected to contribute thinking and strategic recommendations for university management, public policy makers, and organizational actors in formulating effective competency development programs, in order to escalate

work readiness, technology adaptation flexibility, and the competitiveness of the workforce in the digital era.

2. LITERATURE REVIEW

2.1 Digital Literacy

Digital literacy is represented as a comprehensive integration of technical skills, cognitive dimensions, and individual social capabilities that facilitate the search, assessment, engineering, and dissemination of information through the utilization of technology ecosystems in a safe, effective, and purpose-oriented manner. The conceptualization proposed by Reddy et al. (2023) places digital literacy in a multivariate scheme that is specifically designed to cut the disparity in current technological skills, with an emphasis that every individual is required to have critical thinking in analyzing and implementing digital data productively in various academic and professional situations, not just mastering the operational functionality of the device. In line with the premise, Khan et al. (2022) interprets this construct in the scope of higher education as a group of strategic competencies that harmonize the theoretical learning process at the university with the real qualifications of the modern industrial sector, while emphasizing that the group of university graduates with a superior level of technology mastery have a much more mature readiness to navigate the dynamics of changes in the labor market. In a more specific context in Indonesia, Winarno et al. (2024) found empirical evidence that the mastery of digital literacy accompanied by strong self-efficacy is a significant predictor variable for the maturity of work readiness in the group of office administration students, a finding that goes hand in hand with the focus of government macro policies in accelerating the formation of a digitally responsive workforce to support the achievement of the target of massive national digital economy expansion.

2.2 Attitude Towards Digital Literacy for Technology

Attitude towards digital literacy is an evaluative disposition that includes cognitive, affective, and individual behavioral aspects in operating digital technology. In the professional realm, this positive assessment triggers employee motivation to adopt technological innovation to increase work productivity (Nikou et al., 2022). Meanwhile in the university environment, this construct materializes in the form of students' belief in the utility of technology, cyber comfort, and self-efficacy to achieve

learning targets (Getenet et al., 2024). Empirically, positive orientation on information literacy, media, and digital is a strong predictor that consistently affects labor absorption (employability) in Indonesia and Malaysia (Kodrat et al., 2024). In addition, students with a constructive view of digital instruments show massive involvement in online learning, which has a direct impact on the improvement of academic achievement and their readiness to enter the technology-based job market (Getenet et al., 2024).

2.3 Intention to Use Technology

The intention to use technology is the individual's volitional readiness to adopt and integrate digital devices, which based on TAM and TPB act as a proximal predictor of actual utilization behavior. In the work ecosystem, the intensity of this intention is driven by digital literacy capabilities and personal attitude orientation (Nikou et al., 2022). Along with the massive integration of artificial intelligence (AI) in the industry, the intrinsic motivation to explore technology is now transforming into an independent element that accelerates career advancement (Kaya et al., 2025). Empirically, the readiness to interact with the digital ecosystem is crucial in determining the competitive advantage of human resources in the Indonesian and Malaysian labor markets (Kodrat et al., 2024). In the context of universities, students with high determination in implementing AI are proven to be superior in mastering advanced digital skills according to industry standards (Shapiro & Feldman, 2024). Mastery of this competence contributes to the optimization of talent development and career achievement, with the orientation of attitudes towards AI acting as a mediation variable (Kaya et al., 2025).

2.4 Career Adaptability

Career adaptability is an individual's psychosocial capacity to manage transitions, challenges, and vocational tasks based on four dimensions: concern, control, curiosity, and confidence (Super & Savickas in Song et al., 2023). The toughness of this concise measuring instrument of the construct (CAAS-SF) has been methodologically tested on new graduates in China (Song et al., 2023). In employment discourse, this competence is seen as a proactive response of workers to adjust the vocational direction with the fluctuation of industrial market needs (Mahfud et al., 2025). Empirical evidence in Southeast Asia confirms that career adaptability is a crucial capital that bridges the graduation period to the absorption of the world of work (Asok Kumar et

al., 2024). Therefore, the integration of career adaptability training into the vocational curriculum has been proven to significantly boost the mental and operational readiness of graduates to directly enter the industrial ecosystem (Mahfud et al., 2025).

2.5 Employability

Employability (labor absorption capacity) is a multidimensional construct in the form of a combination of dynamic and contextual skills, knowledge, and personal characters to increase job opportunities and career success (Cheng et al., 2022). In the era of digital transformation, the level of work absorption is highly dependent on the suitability of individual digital skill profiles with the technological qualifications needed by the industry (Borsic et al., 2025). Conceptually, this variable is produced by the interaction between career adaptability, digital literacy, and self-efficacy mediated by work readiness (Jaganathan et al., 2025). Empirically, the development of digital skills is proven to have a significant positive impact on employability in all industrial sectors (Borsic et al., 2025). Mastery of information, media, and digital literacy is also a strong predictor of labor superiority in Indonesia and Malaysia (Kodrat et al., 2024). Furthermore, testing based on human capital theory confirms that the ownership of digital employability skills actually boosts the work readiness of vocational education graduates (Oyelaran et al., 2025).

2.6 The Effect of Digital Literacy on Employability

The theoretical foundation of the relationship between digital literacy and employmentability is most strongly placed in the Human Capital Theory which states that the acquisition of digital competence is a form of productive capital that produces measurable returns in labor market participation, where Borsic et al. (2025) rooted their cross-sectoral analysis explicitly in this tradition by arguing that digital skills are a category of human capital that is increasingly irreplaceable in all modern economic sectors; this perspective is reinforced by the Socio-Technical System Theory which states that productive performance in contemporary organizations requires individuals to interact effectively with the technological dimension of work, and Khan et al. (2022) elaborates this relationship by arguing that universities that embed the development of digital literacy in their curriculum directly invest in the human capital of their graduates; empirically, Borsic et al. (2025) found that the achievement of digital skills has a

statistically significant positive effect on employability results where individuals with higher digital competence consistently outperform colleagues who are less.

Hypothesis 1: The level of digital literacy gives a positive and significant contribution directly in increasing the career adaptability of students.

2.7 The Effect of Attitude Towards Digital Literacy for Technology on Employability

The relationship between attitude towards digital literacy and workforce absorption (employability) is rooted in TPB and TAM, which position attitude orientation as the main driver of technology adoption. In TAM, the attitude formed by the perception of usefulness and ease becomes the cognitive-affective mediator of the adoption decision. In the field of work, employees with a positive assessment of digital literacy are intrinsically motivated to use technology to boost performance (Nikou et al., 2022). Meanwhile, through the Human Capital Theory, the disposition of attitude towards digital employability expertise is seen as a psychological capital that forms a comprehensive work readiness (Oyelaran et al., 2025). Empirically, a positive attitude towards information and digital literacy is proven to increase technology adoption and work productivity (Nikou et al., 2022). A comparative study of Indonesia-Malaysia confirms that the dimension of attitude and competence of information literacy, media, and digital consistently predicts employability (Kodrat et al., 2024). PLS-SEM analysis also proves that the positive orientation on digital employability skills is a significant predictor for workforce readiness (Oyelaran et al., 2025). On the basis of the convergence of this evidence, submitted.

Hypothesis 2 : The positive attitude of students towards the utilization of technology has a positive and significant effect directly on the development of their career adaptability.

2.8 The Effect of Intention to Use Technology on Employability

The intention of using technology is the main proximal predictor of actual utilization behavior based on the TAM and TPB framework. In the context of employability, a strong volitional commitment to interact with digital devices encourages individuals to improve their skills and integrate technology into work practices. The orientation of this technology involvement becomes a crucial determinant for individual competitiveness in the digitally transformed labor market

(Kodrat et al., 2024), including the adoption of generative AI by students that has a great impact on their future career trajectory (Shapiro & Feldman, 2024). Empirically, the readiness to engage with digital technology is positively and significantly correlated with the level of employability in Indonesia and Malaysia (Kodrat et al., 2024). Through the PLS-SEM analysis, the mastery of digital employability skills is proven to be a significant predictor of labor readiness (Oyelaran et al., 2025). This is reinforced by evidence that students with high determination to adopt AI are able to develop advanced digital competencies more superiorly according to industry qualifications (Shapiro & Feldman, 2024). On the basis of the scientific convergence, proposed

Hypothesis 3 : The individual's intention or strong intention to adopt digital devices has a positive and significant impact directly on the student's career adaptability capacity.

2.9 The Effect of Digital Literacy on Career Adaptability

The theoretical relationship between digital literacy and career adaptability is explained coherently through Career Construction Theory and Resource Conservation Theory (COR). Digital literacy functions as a cognitive-functional resource that helps individuals collect career information, utilize digital job search tools, and respond flexibly to industry changes. This digital competence equips individuals with informational capital for career planning (Liu et al., 2023), while together with self-awareness building a professional foundation for adaptive career behavior (Soeprijanto et al., 2022). Empirically, mastering digital competence correlates positively and significantly with all dimensions of career adaptability in students in China (Liu et al., 2023). This is reinforced by the findings of mixed methods that show a positive relationship between digital literacy and student career competence (Pan & Bai, 2025). In Indonesia, digital literacy is proven to be a significant predictor for the learning achievement of vocational students which at the same time provides a skill base to develop career adaptability (Soeprijanto et al., 2022). Based on the convergence of this evidence, submitted.

Hypothesis 4 : Digital literacy skills are proven to have a direct positive and significant influence in increasing the absorption index of student labor.

2.10 The Effect of Attitude Towards Digital Literacy for Technology on Career Adaptability

Bandura's (1986) Social Cognitive Theory explains that cognitive and affective orientation guides individual self-regulation in dealing with a dynamic environment. In this context, a positive view of digital literacy acts as a cognitive driver that triggers active involvement in technological changes to support career adaptability. The mechanism is reinforced by Getenet et al. (2024) who found that students' positive attitudes towards digital technology significantly increase digital literacy and self-efficacy, which in turn encourages adaptive responses to academic and professional demands. Meanwhile, Gao and Tadesse (2025) proved that self-efficacy in digital information literacy is a mediator between psychological adaptability and work involvement in students in China. Empirically, the integration of positive attitudes and technology mastery is proven to be directly related to the improvement of competence and individual career flexibility (Pan & Bai, 2025).

Hypothesis 5 : Individual attitudes regarding the usefulness of technology do not show a significant positive impact directly on the achievement of student labor absorption.

2.11 The Effect of Intention to Use Technology on Career Adaptability

The theoretical link between the intention of technology adoption and career adaptability can be explained through the integration of TAM and Career Construction Theory. The intention to use technology (TAM) represents a motivational drive that is in line with the dimension of caring (concern) and curiosity (curiosity) in career development. Conceptually and empirically, (Kumi et al., 2024) proved that technological readiness triggers cognitive flexibility and predicts adaptive work behavior in employees in Malaysia, with career adaptability as the mediator. Relevant to this, Nikou et al. (2022) confirmed that literacy and the tendency to adopt technology are positively correlated with work adjustment capacity. Furthermore, Kaya et al. (2025) expands this scope to the domain of artificial intelligence (AI) and shows that the attitude and intention of technology adoption mediates the influence of digital literacy on individual career growth.

Hypothesis 6 : Behavioral commitment in the form of intention to use digital instruments does not show a significant positive influence directly in stimulating the absorption of student labor.

2.12 The effect of career adaptability on employability

The relationship between career adaptability and work ability (employability) is rooted in Savickas' Career Construction Theory (2013). This theory explains that the four psychosocial dimensions care, control, curiosity, and confidence allow individuals to overcome the work transition and meet industry competence expectations. Conceptually, career adaptability serves as a dynamic resource to manage the psychological challenges of job search (Liew et al., 2024), converting technical knowledge into broader competencies (Ng et al., 2024), and triggering graduate competitiveness through job readiness (Jaganathan et al., 2025). Empirically, Liew et al. (2024) proved a significant positive influence of career adaptability on the perception of employability of the future of new graduates in Malaysia. This positive impact is also confirmed on the work skills of engineering graduates (Ng et al., 2024) and is supported conceptually in the context of higher education in Southeast Asia (Jaganathan et al., 2025).

Hypothesis 7 : The career adaptability of students has direct positive and significant implications in boosting their labor absorption in the industrial market.

2.13 Mediating effects of employability between digital literacy and career adaptability

The role of career adaptability as a mediator between digital literacy and work capability (employability) can be explained through the integration of Career Construction Theory, Human Capital Theory, and resource-based perspective. Digital literacy acts as human capital that stimulates career adaptability, which then forms psychological readiness and competency profiles to increase work competitiveness. Conceptually, Jaganathan et al. (2025) emphasized that digital mastery activates the adaptive capacity of graduates who contribute to employability through the mediation of work readiness. Empirically, Liu et al. (2023) in China proves that technological competence predicts all dimensions of career adaptability, while Chukwuemeka et al. (2024) strengthens this mediation methodology in the context of work integrated learning. Finally, Jaganathan et al. (2025) validates in the Southeast Asian ecosystem that career adaptability along with work readiness mediates the relationship between digital literacy, self efficacy, and employability of graduates.

Hypothesis 8 : Career adaptability plays a significant mediation role in bridging the link between digital literacy and student labor absorption.

2.14 Mediating effects of employability between attitude towards digital literacy for technology and career adaptability

The mechanism of career adaptability as a mediator between attitudes towards digital literacy for technology and work capability (employability) operates through the attitude-behavior-result chain in Theory of Planned Behavior (TPB). A positive attitude towards technology becomes the initial antecedent that builds motivation to develop adaptive competence, which then indirectly optimizes individual work competitiveness through the career adaptation process. Conceptually, Gao and Tadesse (2025) explain that the self-efficacy of digital information literacy mediates the relationship between psychological adaptability and work involvement, while Pan and Bai (2025) assert that the affective response to technology improves vocational skills through the mastery of adaptive skills. Empirically, Gao and Tadesse (2025) confirmed the mediation function of digital self-efficacy as a form of adaptive psychological process that affects work outside. Relevant to that, Kodrat et al. (2024) found a direct relationship between aspects of attitudes in information, media, and digital literacy with employability in Indonesia and Malaysia which indicates the existence of an intermediate role of adaptability. Finally, Pan and Bai (2025) proved to students in China that digital capacity developed from a positive attitude contributes significantly to their competence as well as their career adaptability.

Hypothesis 9 : Career adaptability functions as an intermediate variable that mediates the relationship between the attitude of technology utilization and the result of labor absorption significantly.

2.15 Mediating effects of employability between intention to use technology and career adaptability

The role of career adaptability as a mediator between the intention to use technology and work capability (employability) can be explained through the synthesis of TAM, Career Construction Theory, and Resource Conservation Theory. The intention to utilize technology reflects motivational resources that if actualized will form digital skills and adaptive orientation, which then triggers work competitiveness through flexible individual profiles. Conceptually, Abdul Hamid et al. (2024) stated that technology readiness stimulates adaptive work behavior through the intermediary of career adaptability, while Kaya et al. (2025) places the attitude and intention of adopting

artificial intelligence (AI) as a mediator that connects digital literacy with career achievement. Empirically, (Kumi et al., 2024) proved to employees in Malaysia that technology readiness has a significant impact on adaptive work behavior by being bridged by career adaptability. This finding is completed by Chukwuemeka et al. (2024) which shows the role of digital literacy mediation in the work integrated learning ecosystem, as well as Kaya et al. (2025) which confirms that the intention towards AI mediates the influence of digital literacy on career growth and talent as an independent path to adaptive competence that supports employability.

Hypothesis 10 : Career adaptability acts as an intermediary mechanism that mediates the influence of the intention to use technology on the significant increase in the absorption capacity of student labor.

3. METHODOLOGY

3.1 Research Methodology

This study adopts a quantitative research approach using an online survey questionnaire to examine the relationships between digital literacy, attitude towards use of digital technologies, intention to use, career adaptability, and employability among undergraduate university students. Because this investigation has an exploratory nature, the determination of subject criteria is not limited to certain scientific fields or industrial sectors.

The data networking process is carried out by distributing questionnaire instruments online utilizing various platforms, including academic networks and social media, by appointing a group of undergraduate program students as the main respondent base. In addition, the participants were asked to include basic demographic data such as age classification, gender, and educational background track record to support the course of the descriptive analysis stage.

3.2 Measures and Survey Instruments

All constructs in this investigation are evaluated by utilizing instruments adapted from previous established literature. Digital literacy variables are measured through five questions that represent the respondent's capabilities in learning

technology updates, technical prowess, searching and interpreting information, and caring for cyber issues such as digital security and plagiarism.

Attitude variables towards the use of digital technology are assessed using four questions that capture the respondent's point of view and preference regarding the operation of digital devices in the scope of work and learning. Examples of measurement indicators include the statement that respondents like to operate digital technology to support work or study activities, as well as the statement that the presence of digital technology makes the work or study process more interesting.

Variable intentions in the use of digital technology are estimated through three questions that reflect the willingness and probability of respondent behavior to implement digital devices in searching for information. This indicator includes several statements such as the respondent's plan to operate digital technology in collecting information, as well as the respondent's commitment to continue to utilize digital technology in the future.

Career adaptability variables are measured through eight questions that focus on personal readiness in managing assignments and challenges in the world of work, including aspects of decision-making, professional responsibility, skill improvement, and problem-solving methodology. Examples of operational indicators include activities to learn new skills as well as the ability to solve various work obstacles.

The labor absorption variable (*employability*) is quantified using five questions that net the respondent's perception of their subjective capacity in obtaining and maintaining job positions in the industrial market. Sample statements include “I am confident about my chances of getting a job” and “My qualifications make me competitive in the job market.”

All items were measured using a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). The collected data were analyzed using SmartPLS to assess the relationships between constructs.

3.3 Data Collection and Descriptive Statistics

Data collection was conducted over a period of approximately two months. Survey invitations were distributed to a targeted sample of university students. And get a total 150 respondents.

4. RESULTS AND DISCUSSION

4.1 Measurement Model Assessment

The measurement model assessment was conducted using SmartPLS to evaluate the validity and reliability of the variables used in this study, namely Digital Literacy (DL), Attitude Toward Usage (ATU), Intention to Use (ITU), Career Adaptability (CA), and Employability (EM). The evaluation included convergent validity, discriminant validity, and internal consistency reliability.

4.1.1 Convergent Validity

Convergent validity was assessed using outer loading values and Average Variance Extracted (AVE). According to Hair et al. (2022), indicators with loading values above 0.70 are considered acceptable.

Table 1. Outer Loadings

	ATU	CA	DL	EM	ITU
ATU1	0.821				
ATU2	0.840				
ATU3	0.855				
ATU4	0.790				
CA1		0.747			
CA2		0.766			
CA3		0.789			
CA4		0.716			
CA5		0.765			

CA6		0.746			
CA7		0.778			
CA8		0.811			
DL2			0.775		
DL3			0.809		
DL4			0.834		
DL5			0.843		
EM1				0.826	
EM2				0.804	
EM3				0.724	
EM4				0.763	
EM5				0.862	
ITU1					0.812
ITU2					0.815
ITU3					0.822
DL1			0.800		

All indicators demonstrated the loading values are above the recommended threshold of 0.70, indicating satisfactory convergent validity. These findings suggest that each indicator adequately represents its respective construct.

Table 2. Reliability and Validity Assessment

	Cronbach's alpha	Composite reliability (rho_a)	Composite reliability (rho_c)	Average variance extracted (AVE)

ATU	0.846	0.853	0.896	0.684
CA	0.899	0.903	0.919	0.586
DL	0.871	0.877	0.907	0.660
EM	0.857	0.878	0.897	0.636
ITU	0.749	0.750	0.857	0.666

The results indicate that all variables achieved Cronbach's Alpha and Composite Reliability values above 0.70, confirming satisfactory internal consistency reliability. In addition, all AVE values exceeded 0.50, indicating adequate convergent validity for all constructs.

4.1.2 Discriminant Validity

Discriminant validity was assessed using the Heterotrait-Monotrait Ratio (HTMT). The HTMT values have to be below the recommended threshold of 0.90, to indicate satisfactory discriminant validity.

Table 3. HTMT Results

	Heterotrait-monotrait ratio (HTMT)
CA <-> ATU	0.431
DL <-> ATU	0.455
DL <-> CA	0.423
EM <-> ATU	0.430
EM <-> CA	0.486
EM <-> DL	0.445
ITU <-> ATU	0.642
ITU <-> CA	0.501
ITU <-> DL	0.504
ITU <-> EM	0.487

The HTMT values ranged from 0.423 to 0.642, all below the recommended threshold of 0.90. Therefore, discriminant validity was successfully established, indicating that each construct is empirically distinct from one another.

4.2 Structural Model Assessment

The structural model assessment was conducted to examine the relationships among constructs and evaluate the explanatory power of the proposed model.

4.2.1 Coefficient of Determination (R^2)

The coefficient of determination (R^2) was used to evaluate the explanatory power of the endogenous variables.

Table 4. R-Square Results

	R-square	R-square adjusted
CA	0.246	0.230
EM	0.298	0.279

The results of data analysis confirm that the variables of digital literacy, attitude orientation towards the use of technology, and the intention of using digital devices simultaneously contribute 24.6% in explaining the variance in the career adaptability construct. In addition, the combined contribution of digital literacy variables, attitude orientation towards the use of technology, the intention of using digital devices, along with career adaptability capacity is able to explain the variance in the level of labor absorption (*employability*) of 29,8%.

Even though the acquisition of the R^2 value is in the middle level classification, the structural model construction built in this study still shows an adequate and acceptable explanatory capacity in predicting various externalities related to potential work absorption.

4.2.2 Hypothesis Testing

Statistical analysis to test the hypothesis in this study applies the resampling bootstrapping method through the SmartPLS software. The testing criteria used refers to the significance level of 5 percent. Based on the standard, a hypothesis is stated to be supported or has a significant effect if the statistical number T exceeds the threshold of 1.96 and the gain of the p value is below 0.05.

Table 5. Path Coefficients and Hypothesis Testing

	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	T statistics (O/STDEV)	P values
ATU -> CA	0.175	0.182	0.089	1.967	0.049
ATU -> EM	0.131	0.136	0.081	1.625	0.104
CA -> EM	0.267	0.268	0.092	2.908	0.004
DL -> CA	0.216	0.220	0.080	2.710	0.007
DL -> EM	0.184	0.186	0.083	2.232	0.026
ITU -> CA	0.234	0.234	0.103	2.277	0.023
ITU -> EM	0.142	0.144	0.090	1.567	0.117

The empirical findings of this modeling confirm that Digital Literacy is a determining factor that has been proven to contribute positively and significantly in increasing Career Adaptability as well as Employment. Furthermore, the Attitude to Use variable shows a significant impact in shaping Career Adaptability, but this variable is proven to be unable to make a meaningful contribution directly to the level of Employability. A similar relationship pattern was also found in the Intention to Use variable, which triggered a significant increase in Career Adaptability but did not show any meaningful direct effect on Employability. On the other hand, the results of structural testing indicate that an individual's ability in Career Adaptability plays an important role and has a significant impact directly in affecting their Employability.

4.2.3 Mediation Analysis

This study applies mediation analysis to test the contribution of the Career Adaptability variable as a mediator. These variables are positioned among the influence of Digital Literacy, Attitude Towards Use, Intention to Use, on Work Ability.

Table 6. Specific Indirect Effects

	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)
ATU -> CA -> EM	0.047	0.049	0.030
DL -> CA -> EM	0.058	0.059	0.030

Based on the results of data evaluation, Career Adaptability is proven to play a significant mediation role in two relationship paths, namely the influence of Digital Literacy on Work Ability and the influence of Attitude Towards Use on Work Ability. The results prove that an individual's career adaptation capacity is a crucial channel that connects digital technology mastery with increased work readiness.

4.3 Discussion

This investigation proves empirically that digital literacy skills make a significant positive contribution to the improvement of career adaptability as well as the expansion of labor absorption (*employability*). Groups of individuals who master information technology capabilities in depth show a better level of efficiency when responding to various challenges in the world of work and show more competitive bargaining value in the industrial market. This relationship pattern is in line with a number of previous scientific findings that consistently underline the urgency of strengthening technology literacy as the main foundation in preparing the workforce to face the evolution of the digital corporate landscape.

On the other hand, the results of the analysis show that the orientation of attitudes towards the use of information technology has a great influence in optimizing individual career adaptability, but the psychosocial construct does not trigger a direct impact on the status of labor absorption. This empirical fact indicates that a constructive view of the usefulness of digital instruments contributes strongly to the maturity of flexible work behavior, although it has not been able to boost the probability of obtaining a job position instantly without a bridge of practical competence.

A similar phenomenon is also identified in the variable intention or intention of using information technology devices that has been proven to significantly affect the maturity of career adaptability without giving meaningful direct implications for the fulfillment of work absorption qualifications. This objective finding reinforces the thesis that a person's volitional drive to explore the digital ecosystem has the potential to train their resilience and vocational flexibility, even though the internal intention capital is considered inadequate to intervene in the external achievement of job absorption directly in the eyes of the employer.

Furthermore, career adaptability capacity was found to have very strong positive implications in determining the level of student labor absorption. The validation of this structural relationship strengthens the theoretical scientific argument that the ownership of adaptive psychosocial competence is an absolute requirement to escalate mental readiness as well as the competitiveness of graduates in the midst of the fierce competence of the modern job market.

Through the mediation test, this research provides further evidence that career adaptability plays a crucial role as an intermediary mechanism that channels the positive implications of mastering digital literacy as well as the orientation of the attitude of using technology towards the optimization of labor absorption. Based on this reality, efforts to strengthen digital competence and the formation of positive technology perceptions will ultimately escalate the individual's employability indirectly through the stimulation of career adaptation capacity first.

Overall, the series of findings in this research project presents a valuable theoretical contribution to the expansion of the treasure trove of scientific literature that correlates information technology capabilities, psychosocial flexibility, and work absorption readiness within the scope of higher education institutions and the maturity of the workforce.

5. CONCLUSION

This research is implemented to test the structural model empirically by integrating the dimensions of technological expertise, behavioral tendencies in adopting digital devices, career flexibility, and the labor absorption index in the undergraduate student population. Based on the validation of hypothesis testing results as well as the

completeness of the inferential statistical analysis process, this study formulates a number of crucial scientific conclusions. The first point shows that capability capital in the form of mastering digital literacy is proven to be a significant direct determinant factor, both in strengthening career adaptability capacity and in expanding student absorption opportunities in the industrial field. The mastery of cognitive and technical aspects in this technology ecosystem presents a real competence foundation for individuals to compete competitively.

Second, structural model testing shows a unique finding regarding the role of attitude variables (attitude towards usage) and behavioral intention (intention to use) in the utilization of technology. These two psychosocial constructs are proven to have a very significant direct positive influence in triggering the maturity of student career adaptability. This relationship indicates that a constructive perspective and volitional intention to explore digital technology has succeeded in growing a sense of concern and curiosity of students in preparing for their professional future.

Third, the research results determined that the orientation of the attitude and the intention of using information technology did not show a significant direct positive impact on the achievement of student work absorption. This phenomenon provides a theoretical affirmation that the affective perception and internal motivation of individuals do not immediately transform into the status of job excellence in the eyes of employers if it is not manifested in tactical action or functional adaptation capacity.

Fourth, career adaptability capacity is proven to have a strong direct positive contribution in boosting the probability of student labor absorption. Students who have psychosocial toughness, self-control, and high self-confidence in managing vocational transitions have a much more mature work readiness to be absorbed by the modern industrial sector.

Fifth, the analysis of indirect effects (indirect effects) confirms that career adaptability plays a vital role as an intermediary variable (mediating role) that bridges the influence of digital literacy and technology utilization attitude towards the optimization of student work absorption. Based on these findings, this mediation function shows that the process of converting digital expertise into real job opportunities requires the involvement of adaptive psychosocial mechanisms.

Technology skills and a positive perspective must be cultivated first through the formation of self-regulation capacity, mental resilience, and career flexibility before finally being able to significantly affect the level of competitiveness of students in the labor market.

6. LIMITATION AND FUTURE RESEARCH

This research was organized with the aim of proving empirically the accuracy of the structural model that connects the dimensions of technological proficiency, behavioral orientation in adopting digital devices, career flexibility, and the employment index (employability) in the undergraduate student group. Based on the validation of hypothesis testing results as well as the completeness of the inferential statistical analysis process, this study formulates a number of crucial scientific conclusions. The first point shows that capability capital in the form of mastering digital literacy is proven to be a significant direct determinant factor, both in strengthening career adaptability capacity and in expanding student absorption opportunities in the industrial field. The mastery of cognitive and technical aspects in this technology ecosystem presents a real competence foundation for individuals to compete competitively.

In its development, a comprehensive understanding of digital literacy does not only touch on the operational realm of mechanical devices, but involves the integration of cognitive and social abilities. Through the strengthening of this foundation, final level students are positioned to have a more mature psychological readiness in detecting opportunities as well as transforming industry challenges into career acceleration. When technological instruments transform into catalysts in the modern work environment, this solid literacy capital holding automatically converts academic theoretical potential into functional capabilities that have a high bargain value in front of labor-absorbing organizations.

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